



Research Article

Navigating the complexities of AI and digital governance: the 5W1H framework

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ABSTRACT

As AI and digital technologies advance rapidly, governance frameworks struggle to keep pace with emerging applications and risks. This paper introduces a "5W1H" framework to systematically analyze AI governance proposals through six key questions: *What* should be regulated (data, algorithms, sectors, or risk levels), *Why* regulate (ethics, legal compliance, market failures, or national interests), *Who* should regulate (industry, government, or public stakeholders), *When* regulation should occur (upstream, downstream, or lifecycle approaches), *Where* it should take place (local, national, or international levels), and *How* it should be enacted (hard versus soft regulation). The framework is applied to compare the European Union's AI Act with the current U.S. regulatory landscape, revealing the EU's comprehensive, risk-based approach versus America's fragmented, sector-specific strategy. By providing a structured analytical tool, the 5W1H framework helps policymakers, researchers, and stakeholders navigate complex AI governance decisions and identify areas for improvement in existing regulatory approaches.

In today's digital era, technologies like artificial intelligence (AI), algorithms, and digital currencies have become increasingly ubiquitous. These technologies are advancing so quickly that new applications and use cases often emerge faster than the ethical guidelines and regulatory frameworks needed to govern and regulate them. In response, various stakeholders have proposed different approaches to AI and digital governance, each with their own unique scope, intent, and methodology. By "AI and digital governance," we refer broadly to the policies, procedures, and norms guiding how AI and digital technologies are developed and used (Floridi, 2018). For simplicity, we use "AI governance" and "digital governance" interchangeably and we do not draw a sharp distinction between "governance" and "regulation."¹

The proliferation of AI governance initiatives demonstrates growing global attention to regulation. For example, the European Union passed the AI Act in 2024, establishing the first comprehensive regulatory framework for AI (Future of Life Institute, 2024). The Biden Administration issued an Executive Order on AI in 2023, directing federal agencies to identify the risks of the technology and harness its benefits

(Executive Office of the President, 2023b). In January 2025, the Trump administration revoked this order, with plans to introduce a new AI Executive Order aimed at sustaining and enhancing "America's global AI dominance in order to promote human flourishing, economic competitiveness, and national security" (Executive Office of the President, 2025). Recently, the U.S. House of Representatives tried to pass a budget bill that would have prohibited states from enforcing laws regulating artificial intelligence for the next decade (Hendrix & Lima-Strong, 2025). However, the Senate voted to strip this provision from the bill. Other countries and organizations also have proposed their own national strategies and frameworks for AI regulation. Examples include Canada's Artificial Intelligence and Data Act (AIDA) (Innovation, 2023), the UK's National AI strategy (Department for Science et al., 2021), China's measures to regulate generative AI (Sheehan, 2023), the OECD's AI Principles (OECD, 2019), and the G7 Hiroshima AI Process (OECD, 2023). Beyond government action, leading AI companies such as Anthropic and OpenAI voluntarily committed to promoting the safe, secure, and transparent development of AI technology, including by

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¹ For those interested in exploring how these terms could differ, see Floridi, L. (2018). Soft ethics, the governance of the digital and the General Data Protection Regulation. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 376(2133), 20180081 for a detailed analysis.

establishing frontier AI pre-deployment testing agreements ([Executive Office of the President, 2023a](#)). AlgorithmWatch's Global Inventory for AI Ethics Guidelines now contains more than 160 AI ethics guidelines from companies, governments, and non-profit organizations ([AlgorithmWatch, 2020](#)).

While many people believe that proactive governance is essential, creating effective AI policies remains complex, owing to competing interests and various ethical, legal, and social considerations. Given the rapidly changing nature of these technologies, there is currently no widespread agreement on the best approach to regulate them. Divergence often exists between nations, with some countries taking a more state-centralized approach while others allow private firms to operate with more freedom. Similarly, private firms in the same sector often differ widely in their own approaches to self-regulation.

To navigate this complexity and support comprehensive governance policies, we propose a structured approach to understand and evaluate AI governance proposals. Rather than advocating for any single proposal or making a general prescriptive case for regulation, we break down key decision points to six specific questions, which we call the "5W1H" framework: *What* should be regulated? *Why* should we regulate? *Who* should regulate it? *When* should regulation occur? *Where* should it take place? And, *How* should regulation be enacted? By systematically addressing these questions, we argue that the 5W1H framework provides a conceptual basis for analyzing AI governance proposals and ensures that key aspects are not overlooked. To illustrate this framework's value, we will apply it to two examples: the European Union's AI Act and the current landscape of AI regulations in the United States. We will use the 5W1H framework to identify their strengths and weaknesses and to draw broader lessons for AI governance.

What should be regulated?

In structuring the 5W1H framework, we begin with 'what' should be regulated. This sequence reflects our view that establishing clarity about the object of regulation creates a shared foundation that enables more productive consideration of motivations, authorities, timing, jurisdictions, and methods.

As AI technologies, particularly generative and agentic systems, increasingly rely on vast datasets, it is tempting to craft policies that address data and AI simultaneously. However, distinguishing between data governance and AI regulation can lead to more targeted and effective oversight. One approach then is to focus on the data used to train and develop AI technologies. Data is the foundation for AI capabilities, with different types presenting unique governance challenges. Some data may be personal, such as medical records containing identifiable information, while other data is non-personal. Data structure also varies—it may be structured (clearly labeled in predefined formats), unstructured (like much of the public internet), or semi-structured (falling between these categories). How this data is collected, processed, and used can raise concerns about privacy, provenance, and bias in AI ([Stahl & Wright, 2018](#)). For example, if an AI system is trained on biased data, it may unintentionally reinforce or make decisions based on those biases ([Varona & Suárez, 2022](#)). Similarly, using sensitive personal data without proper protections or proprietary data without proper permissions can violate privacy or intellectual property rights ([Sebastian, 2023](#)). To address these issues, AI developers could be required to ensure that their training data is accurate, representative, and de-identified. They might also need to document the sources of their data thoroughly. Other potential regulations could include stronger privacy protections, such as enforcing strict data security measures and giving users the ability to access, correct, or delete their personal information. In this regard, the European Union's General Data Protection Regulation (GDPR) offers a possible model for this type of AI-specific data governance, with its stringent requirements for data collection, processing, and protection ([Regulation \(EU\) 2016/679, 2016](#)).

A second approach targets the algorithms and AI models themselves.

These include classification and recommendation algorithms powering search engines like Google Search and social media platforms like Facebook, Instagram, and X. They also include newer generative and agentic AI models powering systems such as OpenAI's ChatGPT and Operator, which can produce highly convincing texts and perform actions but may also generate false information ([Hicks et al., 2024](#)) or produce material harms ([Fowler, 2025](#)). The consequences of algorithmic errors can be serious in domains like news reporting, legal decision-making, and medical diagnosis ([Gravel et al., 2023](#)). To mitigate these risks, AI models may require rigorous testing and validation to ensure accuracy, reliability, and safety. Indeed, leading AI labs and government-funded AI safety institutions have already begun implementing various testing protocols to evaluate model performance ([Bengio, 2024b](#)). Pre-deployment safety measures such as adversarial testing ("red teaming") assess their robustness as well as their potential for harm. Post-training evaluations can involve continuous monitoring and updating of models to adapt to new data and emerging challenges. Validation techniques such as cross-validation and A/B testing examine how models perform under extreme or unexpected scenarios and compare model outputs against standards and performance benchmarks. Regulation could help centralize and enforce AI performance standards and benchmarks ([Anderljung et al., 2023](#)). Beyond safety testing and validation, there could also be requirements for algorithmic transparency, which involves documenting the training processes, algorithmic-design choices, and capability evaluations of AI models.² Such transparency could be important for safety and accountability, especially in high-stakes domains such as healthcare or criminal justice. For example, AI systems used in medical diagnoses may need to be able to explain their reasoning in ways that both healthcare professionals as well as patients can trust and understand ([Astromskè et al., 2021](#); [Mittermaier et al., 2023](#)).

A third approach focuses on specific sectors where AI technologies are used. For example, the concerns surrounding AI in self-driving cars are quite different from those regarding automated medical diagnoses or financial decision-making. A sectoral approach would establish tailored rules and guidelines for industries like healthcare, transportation, finance, and education. The U.S. Food and Drug Administration (FDA), for instance, has developed a regulatory framework for AI-based medical devices that includes premarket review, post-market monitoring, and transparency ([U.S. Food & Drug Administration, 2024](#)). However, sectoral AI governance approaches could be limited in their potential to address certain kinds of AI risks or harms, such as private consumer harms or catastrophic risks from AI ([Bengio, 2024a](#)).

Finally, governance could be organized around the level of risk posed by specific AI technologies. The EU's AI Act exemplifies this approach. It distinguishes between unacceptable risk, high risk, limited risk, and minimal risk AI systems, and it imposes more stringent requirements on AI technologies that pose higher risks. Evaluating how risky an AI system and their applications are involves assessing factors such as its intended use, potential for harm, scale of deployment, and the vulnerability of the affected population. Based on the identified risk, suitable regulatory requirements and oversight mechanisms can be put in place. For higher-risk systems and applications, this might include mandatory risk assessments, enhanced transparency and accountability measures, regular audits, and human oversight. In contrast, lower-risk systems and applications could adhere to more limited requirements, such as codes of

² For a detailed exploration of various forms of transparency—such as functional transparency (insight into the high-level algorithmic operation of a system), structural transparency (understanding the implementation of the algorithm in code), and run transparency (comprehending how the program executes on specific hardware and data)—and the challenges inherent in achieving these types of transparency, see [Creel, K. A. \(2020\)](#). Transparency in Complex Computational Systems. *Philosophy of Science*, 87(4), 568-589. <https://doi.org/10.1086/709729>.

conduct or voluntary best practices.

This risk-based approach allows policymakers to focus resources on AI systems posing the greatest societal risks. However, accurately assessing the risk level of AI applications can be difficult, with reasonable disagreements on how to categorize certain use cases (Ebers, 2024). For example, facial recognition technology could be classified differently depending on its application—when used for unlocking smartphones it might be considered low-risk, but when deployed for mass surveillance in public spaces, it could be deemed high-risk. Also, predicting the long-term societal impacts of some AI systems can be challenging. Inaccurate predictions could cause the development of some potentially beneficial AI systems to be prematurely stifled owing to their perceived risks. For instance, increasingly accurate medical AI diagnostic tools such as AI-powered breast cancer detection devices could be prematurely restricted if regulators overestimated their risks, potentially delaying life-saving technologies (Eisemann et al., 2025). Conversely, underestimating risks could result in inadequate regulatory oversight where it is actually needed, such as when algorithmic lending or hiring tools perpetuate historical biases that were not initially recognized during development (Schellmann, 2024).

Why regulate?

Next, we can examine why a certain AI technology or area should be regulated. Many answers can be given, each leading to a different approach to digital governance and regulation.

One reason we might care about regulating AI is because we care about acting morally and ethically. As AI technologies become more advanced and autonomous, they will make decisions that significantly impact our lives, such as in hiring, medical diagnoses, and parole decisions. Indeed, Stephen Hawking and others have cautioned that highly advanced AI could endanger humanity's very survival (Jones 2014). To guide ethical AI development, many experts look to established ethical principles from other fields, such as the four principles of biomedical ethics: autonomy, beneficence, non-maleficence, and justice (Mittelstadt, 2019; Whittlestone et al., 2019). Autonomy requires that AI users be fully informed about its risks and benefits and provide voluntary consent for its use. This is particularly important, as some AI systems can influence or even manipulate our decisions (O'Grady, 2025). Beneficence means ensuring that AI applications prioritize the welfare of users and society, while nonmaleficence requires developers and users to avoid causing unnecessary harm. Justice demands that the benefits and burdens of AI are distributed fairly and equitably.³ Beyond these ethical principles, some other normative considerations include transparency, explainability, and trust, especially since some AI systems operate as "black boxes" not easily understood even by those who program them (Floridi et al., 2018).

It is worth noting that these ethical considerations are not just about avoiding AI's negative impacts. They also provide the moral foundation necessary for digital technologies to contribute meaningfully to positive outcomes, such as the Sustainable Development Goals (SDGs) (Pierson et al., 2025). Indeed, appropriate governance could facilitate sustainable growth and development (Visvizi, 2022), with AI potentially contributing to good health and wellbeing (SDG3), quality education (SDG4), and decent work and economic growth (SDG8) (Meitei et al., 2025). Intelligent decision support systems can also improve political

decision-making, potentially yielding positive societal outcomes across various domains (Reis et al., 2020).

Another key reason to regulate AI is to establish clear legal boundaries and ensure compliance with existing legal frameworks in areas like intellectual property, data protection, tort liability, and health regulations. Without well-defined guardrails, developers and users alike could find themselves in a lawless "wild west," where no one is held accountable for potential harms. However, AI presents unique challenges that may require changes to current laws or the creation of entirely new, AI-specific regulations (Gerke et al., 2020). For example, existing data protection laws may not fully address the issues arising from the vast amounts of data used to train AI models. Also, the ability of some AI systems to make independent decisions could create new problems regarding who is accountable for those decisions.

Some people advocate regulating AI from an economic perspective, such as when market failures occur, in particular, when free markets fail to produce efficient or socially optimal outcomes (Acemoglu, 2023). In theory, perfectly functioning markets would internalize all costs, provide consumers with complete information, and allocate resources efficiently. However, AI markets exhibit certain notable failures. For instance, the economics of AI development seem to favor market concentration, owing to enormous computational requirements, specialized talent needs, and vast data requirements that create significant barriers to entry (Acemoglu, 2023). Without regulation, this concentration could lead to reduced innovation, higher prices, limited access, and the exercise of market power against consumers, workers, and smaller businesses. Also, AI systems can generate significant externalities, effects not reflected in market prices. Negative externalities may include environmental impacts from computing infrastructure, potential job displacement, privacy harms, and societal costs from biased or manipulative systems. Positive externalities may include knowledge spillovers and productivity gains. When market participants do not internalize these costs and benefits, regulation may help address them through taxes, financial incentives, or direct restrictions (Haykel, 2025). However, while the market failure approach provides some justification for regulation, it faces certain issues. For one thing, AI raises questions about dignity, autonomy, and fairness that should not be reduced to economic efficiency considerations (Adriano Delgado Taboada, 2020). Also, identifying and addressing market failures involves value judgments about which outcomes are desirable. This means that economic efficiency alone cannot tell us how to weigh different societal goals or whose interests should take precedence when trade-offs are necessary. Some also question whether regulations can effectively address market failures, given that they evolve slowly compared to rapidly changing technologies (Kevin A. Bryan & Teodoridis, 2024) and that market participants often find creative ways to technically comply while circumventing the regulations' intended purpose.

A further reason to regulate AI may be to boost public trust in AI technologies, which will be crucial if their widespread adoption is warranted. Studies show a decline in public trust in AI companies, owing to concerns about privacy, potential harms, and inadequate testing and evaluation of AI models (Grossman, 2024). In the U.S., trust in AI companies dropped from 50 % in 2019 to 35 % in 2024 (Grossman, 2024). This decline in trust could hinder the adoption of valuable AI applications in healthcare and education. From this pragmatic perspective, AI regulation should focus on implementing measures that enhance public confidence in the safety of AI technologies.

Finally, countries may regulate AI in order to protect their national interests and to position themselves as leaders in AI. AI represents a valuable asset anticipated to confer significant economic, military, and geopolitical advantages (Cave & ÓhÉigeartaigh, 2018). In her analysis of national AI strategies, Roxana Radu notes that countries like the U.S. and the UK explicitly aim for global AI leadership (Radu, 2021). These ambitions are evident in their regulatory approaches. For example, the United States has made substantial investments in AI research and development and has implemented measures to protect American

³ While these principles provide a good starting point, they might need to be part of a more comprehensive ethical framework for effective AI governance. One promising approach is the Fundamental Conditions Approach to human rights developed by S. Matthew Liao (2023). Ethics of AI and Health Care: Towards a Substantive Human Rights Framework. *Topoi*, 42(3), 857-866. This theory posits that human beings have rights to the fundamental conditions needed for pursuing a good life. Liao has argued that this approach can be used to distinguish genuine ethical principles from those that are not.

intellectual property and prevent the transfer of sensitive technologies to foreign adversaries (Radu, 2024). Meanwhile, the UK has adopted a business-friendly and “pro-innovation” approach to AI regulation (Department for Science, 2023). By creating an attractive regulatory environment for AI companies and investors, the UK hopes to position itself as a global hub for AI development and attract global AI talent and investment (Department for Science, 2023; Roberts et al., 2023).

However, prioritizing national interests could lead to a destabilizing AI arms race. If countries put competition ahead of safety, ethics, and international cooperation, they might cut corners or ignore potential risks in their pursuit of cutting-edge AI capabilities. This competitive pressure could lead to the deployment of inadequately tested or poorly regulated AI systems. Moreover, an AI arms race threatens to exacerbate geopolitical tensions (Graylin & Triolo, 2025). Just as the nuclear arms race of the Cold War period led to a proliferation of catastrophic weapons and the constant threat of mutually assured destruction, an unchecked AI race could result in the creation of powerful and uncontrollable AI systems that pose existential risks to humanity (Luckey et al., 2025).

The development of lethal autonomous weapons systems (LAWS) illustrates these dangers. In 2017, a group of leading AI experts and public figures signed an open letter calling for a preemptive ban on LAWS, warning that using these “killer robots” could trigger a “third revolution in warfare,” with potentially devastating consequences for humanity (Future of Life Institute, 2017). Despite their warnings, the development of LAWS has continued, with reports of their use in conflicts such as Libya and Ukraine (Adam, 2024). This situation underscores the urgent need for AI development to be guided by ethical principles, human rights, and the greater good of humanity.

Who should regulate AI?

The question of who should regulate AI involves a range of stakeholders including the government, industry, and the general public. A perspective frequently advanced by technology leaders is self-regulation—the notion that regulation should primarily be managed by those with deep technical expertise. Proponents argue that AI technologies are inherently complex, requiring regulators with strong understanding of their technical capabilities and constraints (Roski et al., 2021). From this viewpoint, policymakers and the general public may lack the specialized knowledge needed to make sound regulatory decisions. Industry self-regulation typically involves companies establishing internal ethics guidelines, technical standards, and best practices without external mandates. This approach has been significant in AI’s early development, with governance measures largely driven by the industry itself. Tech companies often argue that they can respond more nimbly to emerging issues than government regulators, adapting quickly as technologies evolve rather than waiting for slow-moving legislative processes.

However, this approach has limitations. While technical expertise is undoubtedly crucial, it represents just one dimension of effective AI governance. AI systems raise profound ethical, social, and political questions that cannot be reduced to purely technical issues. Consider facial recognition technologies: technical experts can provide valuable insights into the accuracy and limitations of facial recognition algorithms, but the decision of whether and how to deploy these systems is ultimately a normative and political one. It requires weighing competing values such as public safety, privacy, civil liberties, and racial equity. Resolving these trade-offs requires input not just from data scientists, but also from affected communities, ethicists, legal scholars, and public officials.

Moreover, industry self-regulation raises issues relating to conflicts of interest. While many AI companies have formed internal ethics boards and guidelines, these efforts ultimately remain subordinate to corporate financial interests. Meta’s Oversight Board exemplifies this tension (Oversight Board, 2024). Despite its independent adjudicatory function, the board can only affirm or overrule Meta’s original content

moderation decisions without authority to develop alternative remedies or address systemic issues with the company’s business model (Wong & Floridi, 2023). When profits and principles clash, history suggests commercial imperatives can oftentimes prevail, raising questions about whether companies can adequately police themselves on issues affecting public welfare.

At the opposite end of the spectrum lies government-led regulation, with its strongest version advocating full governmental control over digital governance. The idea here is that if the government has all the decision-making power, it can consistently and efficiently guide AI development in ways that serve the public and national interests. However, this approach risks creating an inflexible environment that could stifle innovation. Moreover, putting too much power in the hands of the government may also create opportunities for corrupt or unethical officials to misuse AI technologies or regulatory authority.

Many governance experts emphasize the critical importance of public involvement in AI policy formation (Verhulst, 2018). As AI systems increasingly affect critical domains such as healthcare, education, employment, and criminal justice, it is thought that those directly impacted should have a meaningful input into their development and deployment. Public involvement can happen in various ways, such as through citizen assemblies, discussion forums, public consultations, and participatory design processes (Zahuranec et al., 2019). The aim is to give citizens opportunities to learn about AI, share their values and concerns, and influence the policies and practices that will govern these technologies. This social approach enables AI systems to reflect diverse societal values rather than just the priorities of technologists or policymakers. However, public engagement is not a perfect solution for AI governance challenges. The general public, like policymakers, may lack the technical expertise to fully understand the complexities of AI systems. Also, public participation can sometimes be hijacked or influenced by special interests. To be effective, public engagement processes must therefore be designed to avoid these issues.

Here it is worth mentioning that different regulatory phases may benefit from varying stakeholder engagement levels. Initial agenda-setting and review processes might emphasize broad public consultation, as exemplified by public commentary during rule-making by regulatory authorities like the U.S. Department of Commerce’s Bureau of Industry and Security.⁴ Drafting technical regulations may require deeper expert involvement from industry and academia, while implementation and enforcement benefit from independent oversight bodies that can balance competing interests. Similarly, different regulatory challenges may demand different balances of authority and input. Technical standards might draw heavily on industry expertise, while applications affecting fundamental rights may demand stronger governmental oversight and robust public deliberation.

When should regulation occur?

The timing of AI regulation presents a critical strategic question for policymakers. Three primary approaches have emerged: upstream regulation, downstream regulation, and lifecycle regulation.

Upstream regulation targets the early stages of a technology’s development, establishing boundaries for permissible innovation before widespread deployment. The EU AI Act, for example, bans certain applications like biometric identification systems and manipulative generative AI (European Parliament, 2024). Other measures for upstream regulation might include mandatory impact assessments, ethical review processes, and standards for data collection and use. The main advantage of upstream regulation is that it can help steer AI development in certain, predefined directions and prevent problems before they arise. It establishes clear expectations for developers and can help build public trust in emerging technologies. However, this approach has its

⁴ <https://public-inspection.federalregister.gov/2025-00636.pdf>

challenges. It is hard to predict all the risks and unintended consequences of rapidly evolving AI technologies. Strict regulations early on could stifle innovation and slow down the progress of beneficial applications (Aghion et al., 2023). There's also a risk of regulatory capture, where companies manipulate upstream regulations to block new competitors. One promising avenue for mitigating these drawbacks may be the use of regulatory sandboxes, where prospective regulations are deployed in a limited and controlled manner in order to identify challenges and opportunities before broader implementation (Ruscheimer, 2025).

Downstream regulation focuses on later stages of a technology's lifecycle. Rather than trying to anticipate all potential problems, this approach involves monitoring AI systems that are deployed and managing their effects. Downstream regulation can include performance monitoring, incident reporting, liability and redress mechanisms, and post-deployment risk assessments. In the U.S., both federal government agencies and civil society groups are actively exploring such downstream regulatory interventions (Croxtton et al., 2024). In addition, there are ongoing discussions regarding how conventional accountability mechanisms, such as tort law, might address algorithmic harms and provide compensation to affected individuals (Infantino & Wang, 2019). Other regions have also sought to provide downstream legal protections for individuals. For example, the EU's right-to-be-forgotten allows people to request the removal of inaccurate, irrelevant, or outdated personal information from search engine results (Rosen, 2011).

A strength of downstream regulation is that it allows for more experimentation and innovation during the early stages of AI development. By managing impacts rather than controlling the development processes, this approach enables technological advancement while grounding regulatory responses in actual evidence of harms rather than speculative risks that may never materialize. However, this reactive approach has notable drawbacks. Downstream regulation addresses problems only after they emerge, potentially after harmful AI systems have already achieved widespread deployment. Once a technology becomes deeply embedded in social and economic systems, it becomes politically and practically difficult to regulate effectively (Genus & Stirling, 2018)—just look at the ongoing struggles to regulate social media platforms. In addition, the increasing complexity and opacity of many AI systems makes it hard for regulators to fully understand and assess their impacts after deployment. The rapid proliferation of generative AI tools like ChatGPT has demonstrated how downstream regulation may struggle to effectively address emergent concerns about AI safety and societal disruption once these technologies have been released into the wild (Haque & Li, 2024).

Lifecycle regulation offers a more holistic approach to AI governance by implementing regulatory measures throughout a system's entire existence. In a lifecycle approach, upstream regulations like impact assessments and ethical guidelines are paired with downstream monitoring and adjustments based on real-world performance data. In addition, by establishing continuous feedback loops, knowledge gained at later phases can refine earlier requirements, creating a dynamic, adaptive process. This allows for the maintenance of safety, effectiveness, and ethical standards throughout the product's entire operational lifespan, rather than depending solely on one-time pre-market approvals or post-deployment penalties. According to Vokinger et al., the iterative nature of AI development calls for a lifecycle-based framework to support ongoing product improvement and safety updates (Vokinger et al., 2022). Similarly, Pecoraro and Luzzi argue that integrating risk management processes throughout the lifecycle of medical software can significantly enhance patient and user safety (Pecoraro & Luzzi, 2014).

While lifecycle approaches combine advantages from both upstream and downstream approaches, they also inherit their challenges. Excessive regulation too early risks stifling innovation, while delaying regulatory oversight until after deployment may allow harm to occur before effective intervention can take place. In addition, lifecycle approaches can be more complex and resource-intensive to implement, requiring

sustained coordination among multiple stakeholders and technically complex, real-time adjustments to AI systems. This approach demands greater regulatory capacity and expertise, which may be challenging for jurisdictions with limited resources or technical capabilities.

Effective AI regulation likely requires selective application of upstream, downstream, or lifecycle approaches depending on specific risk profiles, usage contexts, and applications. High-risk applications affecting fundamental rights may warrant stronger upstream controls, while lower-risk applications might benefit from more flexible, downstream approaches.

Regardless of timing strategy, all AI regulatory actions should incorporate evaluation mechanisms. Mandatory review periods or sunset clauses can ensure that policies remain effective and responsive to rapid technological developments. These evaluations should follow predetermined schedules while also allowing for trigger reviews following significant developments such as major technological breakthroughs or safety incidents. To maintain credibility and thoroughness, regulatory reviews should employ comparable rigorous standards used in initial policy development, systematically assessing whether regulations achieve their intended objectives without creating undue constraints on innovation.

Where should regulation take place?

AI regulation can happen at various levels – local, national, and international – or a combination of these approaches. Currently, much of the focus is on national-level regulation. Governments around the world are developing AI strategies and policies that suit their specific economic, social, and political contexts.

National-level AI regulations offer important benefits: they allow countries to tailor AI governance to their unique needs, values, and priorities. For instance, some nations might prioritize rapid innovation and economic growth, while others might emphasize privacy protections or equitable access. National regulations also provide a unified framework to coordinate AI development and deployment across different sectors and regions within a country, creating regulatory certainty for businesses and researchers. In addition, they enable governments to protect their sovereignty and national interests in a domain that has become increasingly important for economic competitiveness, national security, and social stability. However, focusing solely on national regulation has substantial drawbacks. AI technologies and their impacts routinely cross national borders, with data flows, algorithmic decision-making, and digital platforms operating globally. An AI model trained in one country may be deployed in dozens of others, processing data from citizens worldwide and making decisions that affect people across multiple jurisdictions. Divergent national regulations can create a fragmented global landscape, with companies forced to navigate a patchwork of sometimes contradictory rules. This regulatory fragmentation may increase compliance costs, hinder innovation, impede international trade, and complicate collaboration on global challenges. Without international standards, there is also a risk that countries may lower regulatory barriers to attract AI investment and talent, leading to a "race to the bottom" where important safeguards are sacrificed for competitive advantage.

These concerns have prompted calls for increased international cooperation and coordination on AI governance (Ala-Pietilä & Smuha, 2021). Indeed, challenges like data privacy violations, algorithmic bias, AI-generated misinformation, and problematic autonomous system deployments can affect populations worldwide. Similarly, risks such as the concentration of market power in a handful of tech giants (Taeihagh, 2021) or environmentally unsustainable compute requirements can affect all countries (Roberts et al., 2024). Moreover, AI's potential to alter global power structures and widen existing disparities highlights the need for international norms and standards that protect the human rights of vulnerable populations and promote equitable development.

Promising initiatives are underway to create international regulatory

agreements for digital technologies. The EU's AI Act creates a unified framework across 27 Member States, while the G7 countries are developing common principles for responsible AI development and use, through the G7 Hiroshima AI Process (European Commission, 2023).

However, international AI governance also faces significant practical challenges (Roberts et al., 2023). Negotiating binding agreements among nations with differing interests and values is difficult. International diplomacy is slow and often lags behind the rapid pace of AI technology development. Even when agreements are reached, they often lack strong enforcement mechanisms to ensure compliance. Moreover, given the strategic importance of AI for national competitiveness and security, many nations remain reluctant to cede control over their domestic technology policies, complicating international cooperation. Furthermore, international agreements often consist of high-level principles rather than detailed, enforceable regulations. While these principles provide a useful framework for cooperation and norm-setting, they may struggle to address the specific needs and concerns of local communities.

Given these challenges, many advocate for complementary engagement at the local level, involving municipal governments, civil society organizations, academic institutions, and local businesses. New York City's Automated Employment Decision Tool Law, which requires employers to disclose their use of AI in hiring processes, illustrates this "AI localism" approach (Verhulst & Sloane, 2020; Verhulst et al., 2021). Local governance approaches offer several advantages. They can respond more nimbly to community-specific needs and concerns that broader frameworks might overlook, such as how particular AI applications affect marginalized neighborhoods or local industries. Local governance also enables greater public participation by involving residents directly in decisions about technology deployment within their communities. Moreover, local AI governance can function as a policy laboratory, allowing for experimentation and refinement before broader implementation (Marcucci et al., 2022).

At the same time, some limitations of local approaches should be acknowledged. For one thing, local governments may lack the technical expertise and financial resources needed to effectively evaluate sophisticated AI systems, resulting in superficial oversight. Also, a patchwork of inconsistent rules across different jurisdictions increases compliance costs and creates legal uncertainty for companies. This burden can fall disproportionately on startups and smaller businesses that can't afford extensive legal departments to navigate complex regulatory requirements. Lastly, there may be a power imbalance between local governments and tech companies. When municipalities compete for technology investment and jobs, they may face pressure to weaken regulatory standards as companies can credibly threaten to relocate if faced with restrictions they consider too stringent. This economic leverage can seriously undermine effective oversight, especially in smaller communities where limited bargaining power meets strong economic incentives to attract and retain technology businesses.

How should regulations be enacted?

When considering how AI regulations should be enacted, policy-makers should recognize the full spectrum of governance tools available. Governance frameworks typically distinguish between hard regulation and soft regulation, but each encompasses diverse mechanisms that can be strategically deployed to address different aspects of technological governance.

Hard regulation involves legally binding measures that establish mandatory requirements and prohibitions. Rather than merely suggesting best practices, these frameworks create enforceable legal obligations backed by penalties for non-compliance. Hard regulation typically manifests through specific legislation governing areas such as data protection, cybersecurity, and sector-specific applications in healthcare, finance, and transportation. The regulatory framework operates through multiple mechanisms. Legislative bodies create

binding statutes that define requirements and procedures for technology development and deployment. Courts contribute through judicial interpretation, applying existing laws to emerging technological contexts and establishing binding precedents that clarify regulatory boundaries. Regulatory agencies contribute to the enforcement ecosystem by monitoring compliance, investigating violations, and imposing sanctions when organizations fail to meet legal requirements.

The main advantages of hard regulation are its enforceability and consistency. With the full force of law behind it, hard regulation provides stronger protection for individual rights, public values, and policy objectives than voluntary measures. When a practice is legally prohibited, companies face real consequences for violations, creating powerful incentives for compliance. By applying the same rules across all actors in a given jurisdiction, hard regulation also creates a level playing field. It prevents companies from gaining competitive advantages by cutting corners on safety or privacy. It also establishes clear expectations for compliance, reducing uncertainty for businesses regarding what is required of them. For high-risk applications with potential for serious harm, like autonomous vehicles or AI systems making critical healthcare decisions, these legally binding guardrails may be essential for public safety and trust.

However, hard regulation can be slow and less flexible. It can also be costly and burdensome to comply with, especially for small and medium-sized companies. Large corporations can afford dedicated legal teams to navigate complex regulations and streamline compliance, potentially giving them a competitive advantage over smaller rivals. Furthermore, passing such regulations requires significant political will and agreement, which can be challenging in a divided society. Throughout the regulatory process, industry lobbying and political maneuvering may weaken proposed measures or delay their implementation.

Soft regulation, by contrast, typically relies on voluntary measures rather than legally binding rules. This regulatory model operates through several mechanisms. Professional self-regulation represents one primary avenue, including industry codes of conduct, ethical guidelines, and best practices. The proliferation of AI ethics principles among major technology companies exemplifies this trend, with numerous firms publishing voluntary guidelines to govern their AI development activities (AlgorithmWatch, 2020). Technical integration offers another approach through design principles such as "ethics by design" and "privacy by design," which embed governance considerations directly into development processes (Brey & Dainow, 2024). For example, Apple's implementation of privacy-enhancing technologies shows how companies can protect user data through technical safeguards rather than relying exclusively on legal compliance. Soft regulation may also involve creating internal ethics boards, adopting voluntary transparency and accountability measures, and developing industry-wide technical standards and certifications. Lastly, behavioral interventions and market-based incentives—such as reputational benefits for responsible practices—can also function as soft governance mechanisms. While these measures lack legal enforceability, they create shared expectations and behavioral norms that can significantly influence conduct across entire industries or technological ecosystems.

A key advantage of soft regulation is its flexibility and adaptability. Voluntary measures can be developed and updated more quickly than formal laws and regulations. When a new concern emerges, industry groups can develop guidelines or standards to address it without waiting for the hard regulatory legislative process to unfold. Additionally, soft regulation can rely on reputational incentives. Companies often adopt voluntary measures to maintain positive public perceptions and avoid negative publicity, even absent legal requirements. Voluntary certifications or commitments can become powerful competitive differentiators, enabling companies to signal trustworthiness and responsibility to customers, partners, and investors. Perhaps most significantly, soft regulation's voluntary nature makes it more innovation-friendly than hard regulatory approaches. Since companies

retain discretion over implementation methods and can adapt measures to their specific contexts, soft regulation reduces the risk of stifling technological development through overly prescriptive requirements.

However, soft regulation also has limitations. Most obviously, it lacks legal force and enforceability. Companies can violate voluntary commitments with limited legal consequences. Moreover, the high-level principles and standards that different companies espouse often vary across jurisdictions and organizations, and it is not clear whether all of them reflect genuine ethical principles (Liao, 2023). This fragmentation can undermine the effectiveness of soft regulation in establishing clear norms and expectations. Equally as concerning, soft regulation remains vulnerable to corporate capture (Levi-Faur & Mitnick, 2011). Without formal oversight mechanisms, voluntary standards and practices may prioritize industry interests over public welfare. This is a serious concern given the concentration of frontier AI development among a few powerful technology companies with substantial market power and political influence.

In practice, regulating digital technologies often involves a combination of soft and hard measures, depending on what is being regulated. For example, the EU's GDPR is a hard regulation with mandatory requirements for the processing of personal data, but also includes provisions for codes of conduct for flexible and collaborative compliance (Voigt et al., 2017). Similarly, financial regulations often combine legally binding requirements with industry-developed standards and guidance.

Applying the 5W1H framework

To illustrate how the 5W1H framework can be used to analyze and compare AI governance approaches, let's apply it to the EU AI Act and the current landscape of AI regulations in the U.S.

EU's AI Act:

The EU's AI Act is a European, sector-agnostic regulatory framework, and the first comprehensive regulation on artificial intelligence that will apply to all Member States in the European Union. It was first proposed in April 2021 by the European Commission, with the final version being unanimously endorsed by all twenty-seven Member States of the EU in May 2024. The Act's implementation takes a staggered approach, with prohibitions on certain AI systems coming into force in February 2025. Member States must designate national competent authorities by August 2025.⁵

What: As noted earlier, the AI Act uses a risk-based approach to regulate AI, classifying systems into four levels of risk: unacceptable, high, limited, and minimal. Systems with unacceptable risk, like those manipulating human behavior or doing social scoring, are banned (Future of Life Institute, 2024). High-risk systems, like those used in critical infrastructure, aviation, cars, and medical devices, must meet strict requirements for data quality, human oversight, and robustness. Limited risk systems, like chatbots, need to fulfill transparency obligations so users know that they are interacting with AI. Minimal risk systems, such as spam filters, are mostly unregulated but encouraged to follow voluntary codes of conduct.

Why: The AI Act seeks to ensure that AI systems are safe, trustworthy, and aligned with core EU values, including fundamental rights (e.g., privacy, non-discrimination, and human dignity), democracy, and the rule of law. It also aims to enhance the EU's position in the global AI market.

Who: The European AI Office, established within the European Commission, oversees the implementation of the AI Act, working with Member States and external experts. This office has the authority to enforce the rules and impose penalties for non-compliance.

When: The AI Act applies throughout the lifecycle of AI systems, from design and development to deployment. For instance, high-risk

systems must undergo assessments before hitting the market and continuous monitoring afterward to manage any new risks.

Where: The AI Act is binding across all 27 EU Member States, but allows for some national flexibility. Member States must assign specific authorities to enforce the Act or add extra requirements for certain sensitive AI applications within their national context. The Act is also binding overseas insofar as AI companies based outside the EU seek to provide AI products and services to EU Member States.

How: The AI Act primarily relies on hard law mechanisms to achieve its objectives, such as banning non-compliant AI systems from the EU market. It also includes soft regulations, like encouraging voluntary guidelines for AI systems that are not high-risk.

U.S.'s current AI landscape:

What: The U.S. approach to AI regulation remains fragmented, with different sectors subject to varying regulatory frameworks. This piecemeal approach is evident across multiple federal agencies, each applying existing authorities to address AI-related concerns within their jurisdictions. In healthcare, for example, the FDA has issued guidance for AI-based medical devices (U.S. Food & Drug Administration, 2024). The Federal Communications Commission has extended existing telecommunications law to AI applications, ruling that the Telephone Consumer Protection Act's 1990s-era restrictions on "artificial or pre-recorded voice" messages now encompass AI technologies that generate human voices.⁶ Meanwhile, the Federal Trade Commission has warned market participants that using AI tools with discriminatory impacts, making unsubstantiated claims about AI capabilities, or deploying AI systems without adequate risk assessment and mitigation could violate the FTC Act.⁷

Why: U.S. regulatory priorities have shifted significantly between administrations, reflecting different philosophies about the appropriate balance between oversight and innovation. The Biden administration prioritized addressing algorithmic bias, enhancing transparency requirements, and securing frontier AI systems. Legislative proposals such as the Algorithmic Accountability Act exemplified this approach, seeking to prevent AI systems from discriminating against protected groups and establish accountability mechanisms for high-risk applications.⁸ The Trump administration has signaled a more permissive regulatory stance, driven by concerns that excessive oversight could stifle technological progress and undermine U.S. global competitiveness in AI development. This shift reflects a fundamentally different assessment of the trade-offs between regulatory protection and innovation incentives, with the current administration prioritizing market-driven solutions and industry leadership over prescriptive government intervention.

Who: AI regulation in the U.S. involves industry self-regulation, federal agencies, and increasingly, state-level initiatives. On the industry side, leading AI companies have made voluntary commitments toward safe, secure, and transparent AI development (Executive Office of the President, 2023b). At the federal level, key agencies exercise regulatory authority within their respective domains, including the Department of Commerce, the Federal Trade Commission (FTC), and the Food and Drug Administration (FDA). In the absence of comprehensive federal AI legislation, a growing patchwork of state-level AI laws has emerged as individual jurisdictions introduce their own measures to address AI governance gaps (Anderson, 2025).

When: AI regulations in the U.S. have mostly targeted issues after they occur (downstream interventions). For example, the FTC has taken actions against companies misusing facial recognition technology (Schildkraut et al., 2024). There is increasing discussion about addressing issues earlier in the process (upstream measures). The

⁶ <https://docs.fcc.gov/public/attachments/FCC-24-17A1.pdf>

⁷ https://www.ftc.gov/system/files/ftc_gov/pdf/EEOC-CRT-FTC-CFP-B-AI-Joint-Statement%28final%29.pdf

⁸ https://www.wyden.senate.gov/imo/media/doc/algorithmic_accountability_act_of_2023_summary.pdf

⁵ <https://artificialintelligenceact.eu/implementation-timeline/>

Department of Commerce's National Institute of Standards and Technology (NIST), e.g., has developed an AI Risk Management Framework to address potential AI harms throughout the AI lifecycle (Schreck et al., 2023).

Where: Without comprehensive federal legislation, state-level AI regulatory activity has accelerated dramatically, with nearly 700 AI-related legislative proposals under consideration across various states in 2024 alone.⁹ For example, New York City's Local Law 144, passed in December 2021, requires bias audits of AI tools used in employment decisions (BCLP Law, 2024). The California Consumer Privacy Act (CCPA) allows consumers to opt out of businesses' use of automated decision-making technology (BCLP Law, 2024). The state of Colorado has passed the most comprehensive AI legislation in the U.S. to date, requiring high-risk AI systems enacted on or after February 2026 "to use reasonable care to protect consumers from any known or reasonably foreseeable risks of algorithmic discrimination in the high-risk system" (State of Colorado, 2024). Similarly, Utah's AI Policy Act establishes liability for AI use that violates consumer protections and mandates disclosure requirements when individuals interact with AI in regulated occupations (State of Utah, 2024).

How: The U.S. has mainly relied on a mixture of soft regulations like voluntary standards and ethical guidelines at the federal level and hard regulations such as liability standards and transparency requirements at the state level. However, there are growing calls for stronger, enforceable rules at the federal level, especially for high-risk domains.

When comparing the two approaches through the 5W1H lens, we see that the EU has taken a more comprehensive, centralized, and predominantly hard regulatory approach, while the U.S. has, at least so far, pursued a more piecemeal, sector-specific, and largely soft regulatory approach. The EU aims to shape AI development through a risk-based, lifecycle approach, while the U.S. for now has focused on addressing specific AI harms as they arise in particular sectors through downstream measures.

The application of the 5W1H framework to these contrasting approaches yields some insights and potential areas for improvement in both jurisdictions. For the U.S., the framework suggests that there are potential benefits to developing a more comprehensive federal AI strategy. For one, the U.S.'s current regulatory approach to AI is fragmented, which may lead to inconsistent standards, benchmarks, and regulatory controls or result in significant gaps in U.S. AI regulation. Also, as discussed earlier, industry self-regulation can lead to conflicts of interest and inadequate oversight. A more comprehensive regulatory strategy could include implementing upstream governance measures and stronger enforcement mechanisms for high-risk applications.

As the EU moves forward with the AI Act, the 5W1H framework highlights the need for flexibility in adapting to the rapidly changing AI landscape. As noted earlier, overestimating risk could inadvertently hinder innovation while underestimating risk could result in inadequate protection. In addition, the 5W1H framework raises concerns about the potential limitations of a centralized regulatory system, which might not capture diverse perspectives and could lead to blind spots in AI governance. Given the global nature of AI development and deployment, both the EU and the U.S. could benefit from international coordination and harmonization of their governance frameworks. The 5W1H framework could provide a common vocabulary and structure for these efforts, helping to identify similarities and differences in regulatory approaches.

Conclusion

As AI continues to grow and affect all aspects of our lives, it is important that we put in place governance frameworks that are both robust and flexible. The 5W1H framework we have proposed prompts us

to systematically consider these key aspects of AI regulation:

- *What* should the object of regulation be (data, algorithms and AI models, specific sectors and technologies, risk-level)?
- *Why* should we regulate (ethics, law, market, national interests)?
- *Who* should regulate (companies and AI researchers, governments, the general public)?
- *When* should regulation occur (upstream, downstream, lifecycle)?
- *Where* should regulation take place (national, international, local)?
- *How* should regulation be enacted (hard regulation, soft regulation)?

By reminding us to address these questions, the 5W1H framework provides a valuable tool for navigating this complicated policy landscape. It ensures that our digital governance policies are grounded in a clear understanding of our objectives, values, and the potential trade-offs involved in different regulatory approaches. When applied to current governance efforts, such as the EU's AI Act and the developing regulatory environment in the U.S., the 5W1H framework illuminates both the strengths of these initiatives and areas that may require further development. Ultimately, an effective digital governance policy should be able to balance promoting innovation and protecting individual rights, public well-being, and human values. The 5W1H framework makes clear the various levers policymakers, technical experts, and the broader public have to achieve this equilibrium.

CRedit authorship contribution statement

S. Matthew Liao: Conceptualization, Supervision, Writing – original draft, Writing – review & editing. **Iskandar Haykel:** Writing – review & editing. **Katherine Cheung:** Writing – review & editing. **Taylor Matallon:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Acemoglu, D. (2023). Distorted innovation: Does the market get the direction of technology right? *AEA Papers and Proceedings*, 113, 1–28. <https://doi.org/10.1257/pandp.20231000>
- Adam, D. (2024). Lethal AI weapons are here: How can we control them? *Nature*, 629, 521–523.
- Adriano Delgado Taboada, B. (2020). *Regulatory intervention beyond fixing market failures*. <https://www.theregview.org/2020/08/11/delgado-taboada-regulatory-intervention-beyond-fixing-market-failures/>.
- Aghion, P., Bergeaud, A., & Van Reenen, J. (2023). The impact of regulation on innovation. *American Economic Review*, 113(11), 2894–2936.
- Ala-Pietilä, P., & Smuha, N. A. (2021). A framework for global cooperation on artificial intelligence and its governance. In B. Braunschweig, & M. Ghallab (Eds.), *Reflections on artificial intelligence for humanity* (pp. 237–265). Springer.
- AlgorithmWatch. (2020). *AI ethics guidelines global inventory*. Retrieved 8/27 from <https://inventory.algorithmwatch.org/>.
- Anderljung, M., Barnhart, J., Korinek, A., Leung, J., O'Keefe, C., Whittlestone, J., Avin, S., Brundage, M., Bullock, J., Cass-Beggs, D., Chang, B., Collins, T., Fist, T.,

⁹ <https://www.bsa.org/news-events/news/2025-state-ai-wave-building-after-700-bills-in-2024>

- Hadfield, G., Hayes, A., Ho, L., Hooker, S., Horvitz, E., Kolt, N., & Wolf, K. (2023). Frontier AI regulation: Managing emerging risks to public safety. *arXiv*. <https://arxiv.org/abs/2307.03718>.
- Anderson, H. (2025). From California to Kentucky: Tracking the rise of State AI laws in 2025. *White & Case*. <https://www.whitecase.com/insight-alert/california-kentucky-tracking-rise-state-ai-laws-2025>.
- Astromské, K., Pečiūsis, E., & Astromskis, P. (2021). Ethical and legal challenges of informed consent applying artificial intelligence in medical diagnostic consultations. *AI & Society*, 36(2), 509–520. <https://doi.org/10.1007/s00146-020-01008-9>
- BCLP Law. (2024). *Insights: US State-by-State AI legislation snapshot*. BCLP - Client Intelligence <https://www.bclplaw.com/en-US/events-insights-news/us-state-by-state-artificial-intelligence-legislation-snapshot.html>.
- Bengio, Y. (2024a). Government interventions to avert future catastrophic AI risks. *Harvard Data Science Review*, (5) <https://doi.org/10.1162/99608f92.d949f941>
- Bengio, Y. (2024b). *International scientific report on the safety of advanced AI: Interim report*. https://assets.publishing.service.gov.uk/media/6655982fdc15efdddf1a842f/international_scientific_report_on_the_safety_of_advanced_ai_interim_report.pdf.
- Brey, P., & Dainow, B. (2024). Ethics by design for artificial intelligence. *AI and Ethics*, 4(4), 1265–1277. <https://doi.org/10.1007/s43681-023-00330-4>
- Bryan, K. A., & Teodoridis, F. (2024). *Balancing market innovation incentives and regulation in AI: Challenges and opportunities*. <https://www.brookings.edu/articles/balancing-market-innovation-incentives-and-regulation-in-ai-challenges-and-opportunities/>.
- Cave, S., & ÓhÉigeartaigh, S.S. (2018). An AI race for strategic advantage: Rhetoric and risks. Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society.
- Creel, K. A. (2020). Transparency in complex computational systems. *Philosophy of Science*, 87(4), 568–589. <https://doi.org/10.1086/709729>
- Croxtton, J., Robusto, D., Thallam, S., & Calidas, D. (2024). Message incoming: Establish an AI incident reporting system. *Federation of American scientists*. <https://fas.org/publication/establishing-an-ai-incident-reporting-system/>.
- Department for Science, I. a. T.. (2023). *A pro-innovation approach to AI regulation*. GOV.UK.
- Department for Science, I. a. T., Office for Artificial Intelligence, Department for Digital, C., Media & Sport, & Department for Business, E. I. S.. (2021). *National AI strategy*. GOV.UK <https://www.gov.uk/government/publications/national-ai-strategy>.
- Ebers, M. (2024). Truly risk-based regulation of artificial intelligence - how to implement the EU's AI Act. SSRN. <https://doi.org/10.2139/ssrn.4870387>
- Eisemann, N., Bunk, S., Mukama, T., Baltus, H., Elsnér, S. A., Gomille, T., Hecht, G., Heywang-Köbrunner, S., Rathmann, R., Siegmann-Luz, K., Töllner, T., Vomweg, T. W., Leibig, C., & Katalinic, A. (2025). Nationwide real-world implementation of AI for cancer detection in population-based mammography screening. *Nature Medicine*, 31(3), 917–924. <https://doi.org/10.1038/s41591-024-03408-6>
- European Commission. (2023). *G7 leaders' statement on the Hiroshima AI process*. European Commission. <https://digital-strategy.ec.europa.eu/en>.
- European Parliament. (2024). *Artificial intelligence act*. Retrieved from https://www.europarl.europa.eu/doceo/document/TA-9-2024-0138_EN.pdf.
- Executive Office of the President. (2023a). Fact sheet: Biden-Harris administration secures voluntary commitments from leading artificial intelligence companies to manage the risks posed by AI. In: The White House.
- Executive Office of the President. (2023b). *Safe, secure, and trustworthy development and use of artificial intelligence*. Retrieved from <https://www.federalregister.gov/documents/2023/11/01/2023-24283/safe-secure-and-trustworthy-development-and-use-of-artificial-intelligence>.
- Executive Office of the President. (2025). *Removing barriers to American leadership in artificial intelligence*. Retrieved from <https://www.federalregister.gov/documents/2023/11/01/2023-24283/safe-secure-and-trustworthy-development-and-use-of-artificial-intelligence>.
- Floridi, L., Cows, J., Beltrametti, M., Chatila, R., Chazerand, P., Dignum, V., Luetge, C., Madelin, R., Pagallo, U., & Rossi, F. (2018). AI4People—An ethical framework for a good AI society: Opportunities, risks, principles, and recommendations. *Minds and machines*, 28, 689–707.
- Floridi, L. (2018). Soft ethics, the governance of the digital and the General Data Protection Regulation. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 376(2133), Article 20180081.
- Fowler, G. A. (2025). I let ChatGPT's new 'agent' manage my life. It spent \$31 on a dozen eggs. *The Washington Post*. <https://img.washingtonpost.com/technology/2025/02/07/openai-operator-ai-agent-chatgpt/>.
- Future of Life Institute. (2017). *An open letter to the United Nations convention on certain conventional weapons*. Future of Life Institute.
- Future of Life Institute. (2024). *The EU artificial intelligence act: Up-to-date developments and analyses of the EU AI Act*. Retrieved July from <https://artificialintelligenceact.eu/>.
- Genus, A., & Stirling, A. (2018). Collingridge and the dilemma of control: Towards responsible and accountable innovation. *Research Policy*, 47(1), 61–69. <https://doi.org/10.1016/j.respol.2017.09.012>
- Gerke, S., Minssen, T., & Cohen, G. (2020). Ethical and legal challenges of artificial intelligence-driven healthcare. In A. Bohr, & K. Memarzadeh (Eds.), *Artificial intelligence in healthcare* (pp. 295–336). Academic Press. <https://doi.org/10.1016/B978-0-12-818438-7.00012-5>.
- Gravel, J., D'Amours-Gravel, M., & Osmanliu, E. (2023). Learning to fake it: Limited responses and fabricated references provided by ChatGPT for medical questions. *Mayo Clinic Proceedings: Digital Health*, 1(3), 226–234.
- Graylin, A. W., & Triolo, P. (2025). There can be no winners in a US-China AI arms race. *MIT Technology Review*. <https://www.technologyreview.com/2025/01/21/1110269/there-can-be-no-winners-in-a-us-china-ai-arms-race/>.
- Grossman, G. (2024). *Rebuilding trust to reach AI's potential*. Edelman. <https://www.edelman.com/insights/rebuilding-trust-reach-ai-potential>.
- Haque, M. A., & Li, S. (2024). Exploring ChatGPT and its impact on society. *AI and Ethics*, 1–13.
- Haykel, I. (2025, March 27, 2025). AI security tax incentives. *Americans for responsible innovation*. <https://ari.us/policy-bytes/ai-security-tax-incentives/>.
- Hendrix, J., & Lima-Strong, C. (2025). *US House passes 10-Year moratorium on state AI laws*. May 22, 2025. Tech Policy Press <https://www.techpolicy.press/us-house-passes-10-year-moratorium-on-state-ai-laws/>.
- Hicks, M. T., Humphries, J., & Slater, J. (2024). ChatGPT is bullshit. *Ethics and Information Technology*, 26(2), 38.
- Infantino, M., & Wang, W. (2019). Algorithmic torts: A prospective comparative overview. *Transnational Law & Contemporary Problems*, 28(2).
- Innovation, Science, and Economic Development Canada. (2023). *The Artificial Intelligence and Data Act (AIDA) – Companion document*. In.
- Levi-Faur, D., & Mitnick, B. M. (2011). *Handbook on the politics of regulation*. Edward Elgar Publishing. <https://doi.org/10.4337/9780857936110>
- Liao, S. M. (2023). Ethics of AI and health care: Towards a substantive Human rights framework. *Topoi*, 42(3), 857–866. <https://doi.org/10.1007/s11245-023-09911-8>
- Luckey, D., Muggy, S. D., Frey, T., Stebbins, D., Rissman, T., Espinosa, B., Tapia, D., Greg Mckelvey, J., Pokhriyal, N., Dawson, J., Hughes, S., Sandler, M., Bakhshi, R., Kepe, M., Kirkwood, G., Denton, S. W., Desmet, D., Makino, M., Guest, E., & Ryseff, J. (2025). *Mitigating risks at the intersection of artificial intelligence and chemical and biological weapons*. https://www.rand.org/pubs/research_reports/RRA2990-1.html.
- Marcucci, S., Kalkar, U., & Verhulst, S. (2022). *AI localism in practice: Examining how cities govern AI*. <https://ssrn.com/abstract=4284013>.
- Meitei, A. J., Rai, P., & Rajkishan, S. S. (2025). Application of AI/ML techniques in achieving SDGs: A bibliometric study. *Environment, Development and Sustainability*, 27(1), 281–317. <https://doi.org/10.1007/s10668-023-03935-1>
- Mittelstadt, B. (2019). Principles alone cannot guarantee ethical AI. *Nature Machine Intelligence*, 1(11), 501–507. <https://doi.org/10.1038/s42256-019-0114-4>
- Mittermaier, M., Raza, M. M., & Kvedar, J. C. (2023). Bias in AI-based models for medical applications: Challenges and mitigation strategies. *npj Digital Medicine*, 6(1), 113. <https://doi.org/10.1038/s41746-023-00858-z>
- O'Grady, C. (2025). Unethical AI research on Reddit under fire. *Science (New York, N. Y.)*, 388(6747), 570–571. <https://www.science.org/content/article/unethical-ai-research-reddit-under-fire>.
- OECD. (2019). *OECD AI principles overview*. OECD.AI.
- OECD. (2023). *G7 Hiroshima process on generative artificial intelligence (AI): Towards a G7 common understanding on generative AI*. OECD iLibrary.
- Oversight Board. (2024). Improving how meta treats people and communities around the world. In.
- Pecoraro, F., & Luzi, D. (2014). The integration of the risk management process with the lifecycle of medical device software. *Methods of information in medicine*, 53(2), 92–98. <https://doi.org/10.3414/me13-01-0014>
- Pierson, E., Shanmugam, D., Movva, R., Kleinberg, J., Agrawal, M., Dredze, M., Ferryman, K., Gichoya, J. W., Jurafsky, D., Koh, P. W., Levy, K., Mullainathan, S., Obermeyer, Z., Suresh, H., & Vafa, K. (2025). Using large language models to promote health equity. *NEJM AI*, 2(2), Article Alp2400889. <https://doi.org/10.1056/AIp2400889>
- Radu, R. (2021). Steering the governance of artificial intelligence: National strategies in perspective. *Policy and Society*, 40(2), 178–193.
- Radu, R. (2024). *The variable geometry of AI governance*. Geneva Policy Outlook. <https://www.genevapolicyoutlook.ch/the-variable-geometry-of-ai-governance/>.
- Regulation (EU). 2016/679. European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation) (Text with EEA relevance), 1-88 119 (2016). <http://data.europa.eu/eli/reg/2016/679/oj>.
- Reis, J., Santo, P., & Melão, N. (2020). Impact of artificial intelligence research on politics of the European Union member States: The case study of Portugal. *Sustainability*, 12(17), 6708. <https://www.mdpi.com/2071-1050/12/17/6708>.
- Roberts, H., Babuta, A., Morley, J., Thomas, C., Taddeo, M., & Floridi, L. (2023). Artificial intelligence regulation in the United Kingdom: A path to good governance and global leadership? *Internet Policy Review*, 12(2). <https://doi.org/10.14763/2023.2.1709>
- Roberts, H., Hine, E., Taddeo, M., & Floridi, L. (2024). Global AI governance: Barriers and pathways forward. *International Affairs*, 100(3), 1275–1286. <https://doi.org/10.1093/ia/iaae073>
- Rosen, J. (2011). The right to be forgotten. *Stanford Law Review Online*, 64, 88.
- Roski, J., Maier, E. J., Vigilante, K., Kane, E. A., & Matheny, M. E. (2021). Enhancing trust in AI through industry self-governance. *Journal of the American Medical Informatics Association*, 28(7), 1582–1590.
- Ruschmeier, H. (2025). *Thinking outside the box? Bridging the gap between AI and reality*. Cham.
- Schellmann, H. (2024). *The algorithm*. Grand Central Publishing.
- Schildkraut, P.J., Sabet, A., & Perkins, N.L. (2024). The FTC cracks down on Rite Aid's deployment of AI-based technology. Arnold & Porter. <https://www.arnoldporter.com/en/perspectives/blogs/consumer-products-and-retail-navigator/2024/01/the-ftc-cracks-down-on-rite-aid>.
- Schreck, M., Gomez, M., & Charkoudian, S. G. (2023). *US Artificial Intelligence Regulations: Watch list for 2023: An overview of the landscape for US regulation of AI technology*. https://www.goodwinlaw.com/en/insights/publications/2023/04/04_12-us-artificial-intelligence-regulations.

- Sebastian, G. (2023). Privacy and data protection in ChatGPT and other AI chatbots: Strategies for securing user information. *International Journal of Security and Privacy in Pervasive Computing (IJSPPC)*, 15(1), 1–14.
- Sheehan, M. (2023). China's AI regulations and how they get made. *Horizons: Journal of International Relations and Sustainable Development*, (24), 108–125.
- Stahl, B. C., & Wright, D. (2018). Ethics and privacy in AI and big data: Implementing responsible research and innovation. *IEEE Security & Privacy*, 16(3), 26–33.
- State of Colorado. (2024). *Consumer protections for artificial intelligence*.
- State of Utah. (2024). *Artificial intelligence amendments*.
- Taeihagh, A. (2021). Governance of artificial intelligence. *Policy and Society*, 40(2), 137–157.
- U.S. Food and Drug Administration. (2024). *Artificial intelligence and machine learning in software as a medical device*. U.S. Food and Drug Administration.
- Varona, D., & Suárez, J. L. (2022). Discrimination, bias, fairness, and trustworthy AI. *Applied Sciences*, 12(12), 5826.
- Verhulst, S., & Sloane, M. (2020). *Realizing the potential of AI localism*. Project Syndicate <https://www.project-syndicate.org/commentary/local-regulation-of-artificial-intelligence-uses-by-stefaan-g-verhulst-1-and-mona-sloane-2020-02>.
- Verhulst, S., Young, A., & Sloane, M. (2021). The AI localism canvas. *Informationen zur Raumentwicklung*, 48(3), 86–89.
- Verhulst, S. (2018). Where and when AI and CI meet: Exploring the intersection of artificial and collective intelligence towards the goal of innovating how we govern. *AI & society*, 33, 293–297.
- Visvizi, A. (2022). Artificial Intelligence (AI) and sustainable Development Goals (SDGs): Exploring the impact of AI on politics and society. *Sustainability*, 14(3), 1730. <https://www.mdpi.com/2071-1050/14/3/1730>.
- Voigt, P., von dem Bussche, A., Voigt, P., & von dem Bussche, A. (2017). Practical implementation of the requirements under the GDPR. *The EU general data protection regulation (GDPR)* (pp. 245–249). Springer.
- Vokinger, K. N., Hwang, T. J., & Kesselheim, A. S. (2022). Lifecycle regulation and evaluation of artificial intelligence and machine learning-based medical devices. In I. G. Cohen, T. Minssen, W. N. Price II, C. Robertson, & C. Shachar (Eds.), *The future of medical device regulation: Innovation and protection* (pp. 13–21). Cambridge University Press. <https://doi.org/10.1017/9781108975452.002>.
- Whittlestone, J., Nyrop, R., Alexandrova, A., Dihal, K., & Cave, S. (2019). *Ethical and societal implications of algorithms, data, and artificial intelligence: A roadmap for research*.
- Wong, D., & Floridi, L. (2023). Meta's Oversight Board: A review and critical assessment. *Minds and machines*, 33(2), 261–284. <https://doi.org/10.1007/s11023-022-09613-x>
- Zahuranec, A., Young, A., & Verhulst, S. (2019). *Identifying citizens' Needs by combining artificial intelligence (AI) and collective intelligence (CI)*. https://thegovlab.org/static/files/publications/CI-AI_oct2019.pdf.